

Advanced Panel Data Methods

In this chapter, we focus on two methods for estimating unobserved effects panel data models that are at least as common as first differencing. Although these methods are somewhat harder to describe and implement, several econometrics packages support them.

In Section 14.1, we discuss the fixed effects estimator, which, like first differencing, uses a transformation to remove the unobserved effect a_i prior to estimation. Any time-constant explanatory variables are removed along with a_i .

The random effects estimator in Section 14.2 is attractive when we think the unobserved effect is uncorrelated with all the explanatory variables. If we have good controls in our equation, we might believe that any leftover neglected heterogeneity only induces serial correlation in the composite error term, but it does not cause correlation between the composite errors and the explanatory variables. Estimation of random effects models by generalized least squares is fairly easy and is routinely done by many econometrics packages.

Section 14.3 introduces the relatively new correlated random effects approach, which provides a synthesis of fixed effects and random effects methods, and has been shown to be practically very useful.

In Section 14.4, we show how panel data methods can be applied to other data structures, including matched pairs and cluster samples.

14.1 Fixed Effects Estimation

First differencing is just one of the many ways to eliminate the fixed effect, a_i . An alternative method, which works better under certain assumptions, is called the **fixed effects transformation**. To see what this method involves, consider a model with a single explanatory variable: for each i ,

$$y_{it} = \beta_1 x_{it} + a_i + u_{it}, \quad t = 1, 2, \dots, T. \quad [14.1]$$

Now, for each i , average this equation over time. We get

$$\bar{y}_i = \beta_1 \bar{x}_i + a_i + \bar{u}_i, \quad [14.2]$$

where $\bar{y}_i = T^{-1} \sum_{t=1}^T y_{it}$, and so on. Because a_i is fixed over time, it appears in both (14.1) and (14.2). If we subtract (14.2) from (14.1) for each t , we wind up with

$$y_{it} - \bar{y}_i = \beta_1(x_{it} - \bar{x}_i) + u_{it} - \bar{u}_i, \quad t = 1, 2, \dots, T,$$

or

$$\ddot{y}_{it} = \beta_1 \ddot{x}_{it} + \ddot{u}_{it}, \quad t = 1, 2, \dots, T, \quad [14.3]$$

where $\ddot{y}_{it} = y_{it} - \bar{y}_i$ is the **time-demeaned data** on y , and similarly for $\ddot{x}_{it}$ and $\ddot{u}_{it}$. The fixed effects transformation is also called the **within transformation**. The important thing about equation (14.3) is that the unobserved effect, a_i , has disappeared. This suggests that we should estimate (14.3) by pooled OLS. A pooled OLS estimator that is based on the time-demeaned variables is called the **fixed effects estimator** or the **within estimator**. The latter name comes from the fact that OLS on (14.3) uses the time variation in y and x *within* each cross-sectional observation.

The *between estimator* is obtained as the OLS estimator on the cross-sectional equation (14.2) (where we include an intercept, β_0): we use the time averages for both y and x and then run a cross-sectional regression. We will not study the between estimator in detail because it is biased when a_i is correlated with $\bar{x}_i$ (see Problem 2). If we think a_i is uncorrelated with x_{it} , it is better to use the random effects estimator, which we cover in Section 14.2. The between estimator ignores important information on how the variables change over time.

Adding more explanatory variables to the equation causes few changes. The original **unobserved effects model** is

$$y_{it} = \beta_1 x_{it1} + \beta_2 x_{it2} + \dots + \beta_k x_{itk} + a_i + u_{it}, \quad t = 1, 2, \dots, T. \quad [14.4]$$

We simply use the time-demeaning on each explanatory variable—including things like time-period dummies—and then do a pooled OLS regression using all time-demeaned variables. The general time-demeaned equation for each i is

$$\ddot{y}_{it} = \beta_1 \ddot{x}_{it1} + \beta_2 \ddot{x}_{it2} + \dots + \beta_k \ddot{x}_{itk} + \ddot{u}_{it}, \quad t = 1, 2, \dots, T, \quad [14.5]$$

which we estimate by pooled OLS.

Under a strict exogeneity assumption on the explanatory variables, the fixed effects estimator is unbiased: roughly, the idiosyncratic error u_{it} should be uncorrelated with each explanatory variable across *all* time periods. (See the chapter appendix for precise statements of the assumptions.) The fixed effects estimator allows for arbitrary correlation between a_i and the explanatory variables in any time period, just as with first differencing. Because of this, any explanatory variable that is constant over time for all i gets swept away by the fixed effects transformation: $\ddot{x}_{it} = 0$ for all i and t , if x_{it} is constant across t . Therefore, we cannot include variables such as gender or a city's distance from a river.

EXPLORING FURTHER 14.1

Suppose that in a family savings equation, for the years 1990, 1991, and 1992, we let $kids_{it}$ denote the number of children in family i for year t . If the number of kids is constant over this three-year period for most families in the sample, what problems might this cause for estimating the effect that the number of kids has on savings?

The other assumptions needed for a straight OLS analysis to be valid are that the errors u_{it} are homoskedastic and serially uncorrelated (across t); see the appendix to this chapter.

There is one subtle point in determining the degrees of freedom for the fixed effects estimator. When we estimate the time-demeaned equation (14.5) by pooled OLS, we have NT total observations and k independent variables. [Notice that there is no intercept in (14.5); it is eliminated by the fixed effects transformation.] Therefore, we should apparently have $NT - k$ degrees of freedom. This calculation is incorrect. For each cross-sectional observation i , we lose one df because of the time-demeaning. In other words, for each i , the demeaned errors $\ddot{u}_{it}$ add up to zero when summed across t , so we lose one degree of freedom. (There is no such constraint on the original idiosyncratic errors u_{it} .) Therefore, the appropriate degrees of freedom is $df = NT - N - k = N(T - 1) - k$. Fortunately, modern regression packages that have a fixed effects estimation feature properly compute the df . But if we have to do the time-demeaning and the estimation by pooled OLS ourselves, we need to correct the standard errors and test statistics.

EXAMPLE 14.1

EFFECT OF JOB TRAINING ON FIRM SCRAP RATES

We use the data for three years, 1987, 1988, and 1989, on the 54 firms that reported scrap rates in each year. No firms received grants prior to 1988; in 1988, 19 firms received grants; in 1989, 10 different firms received grants. Therefore, we must also allow for the possibility that the additional job training in 1988 made workers more productive in 1989. This is easily done by including a lagged value of the grant indicator. We also include year dummies for 1988 and 1989. The results are given in Table 14.1.

We have reported the results in a way that emphasizes the need to interpret the estimates in light of the unobserved effects model, (14.4). We are explicitly controlling for the unobserved, time-constant effects in a_i . The time-demeaning allows us to estimate the β_j , but (14.5) is not the best equation for interpreting the estimates.

Interestingly, the estimated lagged effect of the training grant is substantially larger than the contemporaneous effect: job training has an effect at least one year later. Because the dependent variable is in logarithmic form, obtaining a grant in 1988 is predicted to lower the firm scrap rate in 1989 by about 34.4% [$\exp(-.422) - 1 \approx -.344$]; the coefficient on $grant_{-1}$ is significant at the 5% level against a two-sided alternative. The coefficient on $grant$ is significant at the 10% level, and the size of the coefficient is hardly trivial. Notice the df is obtained as $N(T - 1) - k = 54(3 - 1) - 4 = 104$.

The coefficient on $d89$ indicates that the scrap rate was substantially lower in 1989 than in the base year, 1987, even in the absence of job training grants. Thus, it is important to allow for these aggregate effects. If we omitted the year dummies, the secular

increase in worker productivity would be attributed to the job training grants. Table 14.1 shows that, even after controlling for aggregate trends in productivity, the job training grants had a large estimated effect.

Finally, it is crucial to allow for the lagged effect in the model. If we omit

EXPLORING FURTHER 14.2

Under the Michigan program, if a firm received a grant in one year, it was not eligible for a grant the following year. What does this imply about the correlation between $grant$ and $grant_{-1}$?

TABLE 14.1 Fixed Effects Estimation of the Scrap Rate Equation

Dependent Variable: $\log(\text{scrap})$	
Independent Variables	Coefficient (Standard Error)
<i>d88</i>	−.080 (.109)
<i>d89</i>	−.247 (.133)
<i>grant</i>	−.252 (.151)
<i>grant</i> _{−1}	−.422 (.210)
Observations	162
Degrees of freedom	104
<i>R</i> -squared	.201

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*grant*_{−1}, then we are assuming that the effect of job training does not last into the next year. The estimate on *grant* when we drop *grant*_{−1} is −.082 ($t = -.65$); this is much smaller and statistically insignificant.

When estimating an unobserved effects model by fixed effects, it is not clear how we should compute a goodness-of-fit measure. The *R*-squared given in Table 14.1 is based on the within transformation: it is the *R*-squared obtained from estimating (14.5). Thus, it is interpreted as the amount of time variation in the y_{it} that is explained by the time variation in the explanatory variables. Other ways of computing *R*-squared are possible, one of which we discuss later.

Although time-constant variables cannot be included by themselves in a fixed effects model, they *can* be interacted with variables that change over time and, in particular, with year dummy variables. For example, in a wage equation where education is constant over time for each individual in our sample, we can interact education with each year dummy to see how the return to education has changed over time. But we cannot use fixed effects to estimate the return to education in the base period, which means we cannot estimate the return to education in any period; we can only see how the return to education in each year differs from that in the base period. Section 14.3 describes an approach that allows coefficients on time-constant variables to be estimated while preserving the fixed effects nature of the analysis.

When we include a full set of year dummies—that is, year dummies for all years but the first—we cannot estimate the effect of any variable whose *change* across time is constant. An example is years of experience in a panel data set where each person works in every year, so that experience always increases by one in each year, for every person in the sample. The presence of a_i accounts for differences across people in their years of experience in the initial time period. But then the effect of a one-year increase in experience cannot be distinguished from the aggregate time effects (because experience increases by the same amount for everyone). This would also be true if, in place of separate year dummies, we used a linear time trend: for each person, experience cannot be distinguished from a linear trend.

EXAMPLE 14.2

HAS THE RETURN TO EDUCATION CHANGED OVER TIME?

The data in WAGEPAN.RAW are from Vella and Verbeek (1998). Each of the 545 men in the sample worked in every year from 1980 through 1987. Some variables in the data set change over time: experience, marital status, and union status are the three important ones. Other variables do not change: race and education are the key examples. If we use fixed effects (or first differencing), we cannot include race, education, or experience in the equation. However, we can include interactions of *educ* with year dummies for 1981 through 1987 to test whether the return to education was constant over this time period. We use $\log(\text{wage})$ as the dependent variable, dummy variables for marital and union status, a full set of year dummies, and the interaction terms $d81 \cdot \text{educ}$, $d82 \cdot \text{educ}$, ..., $d87 \cdot \text{educ}$.

The estimates on these interaction terms are all positive, and they generally get larger for more recent years. The largest coefficient of .030 is on $d87 \cdot \text{educ}$, with $t = 2.48$. In other words, the return to education is estimated to be about 3 percentage points larger in 1987 than in the base year, 1980. (We do not have an estimate of the return to education in the base year for the reasons given earlier.) The other significant interaction term is $d86 \cdot \text{educ}$ (coefficient = .027, $t = 2.23$). The estimates on the earlier years are smaller and insignificant at the 5% level against a two-sided alternative. If we do a joint F test for significance of all seven interaction terms, we get $p\text{-value} = .28$: this gives an example where a set of variables is jointly insignificant even though some variables are individually significant. [The df for the F test are 7 and 3,799; the second of these comes from $N(T - 1) - k = 545(8 - 1) - 16 = 3,799$.] Generally, the results are consistent with an increase in the return to education over this period.

The Dummy Variable Regression

A traditional view of the fixed effects approach is to assume that the unobserved effect, a_i , is a parameter to be estimated for each i . Thus, in equation (14.4), a_i is the intercept for person i (or firm i , city i , and so on) that is to be estimated along with the β_j . (Clearly, we cannot do this with a single cross section: there would be $N + k$ parameters to estimate with only N observations. We need at least two time periods.) The way we estimate an intercept for each i is to put in a dummy variable for each cross-sectional observation, along with the explanatory variables (and probably dummy variables for each time period). This method is usually called the **dummy variable regression**. Even when N is not very large (say, $N = 54$ as in Example 14.1), this results in many explanatory variables—in most cases, too many to explicitly carry out the regression. Thus, the dummy variable method is not very practical for panel data sets with many cross-sectional observations.

Nevertheless, the dummy variable regression has some interesting features. Most importantly, it gives us *exactly* the same estimates of the β_j that we would obtain from the regression on time-demeaned data, and the standard errors and other major statistics are identical. Therefore, the fixed effects estimator can be obtained by the dummy variable regression. One benefit of the dummy variable regression is that it properly computes the degrees of freedom directly. This is a minor advantage now that many econometrics packages have programmed fixed effects options.

The R -squared from the dummy variable regression is usually rather high. This occurs because we are including a dummy variable for each cross-sectional unit, which explains much of the variation in the data. For example, if we estimate the unobserved effects

model in Example 13.8 by fixed effects using the dummy variable regression (which is possible with $N = 22$), then $R^2 = .933$. We should not get too excited about this large R -squared: it is not surprising that we can explain much of the variation in unemployment claims using both year and city dummies. Just as in Example 13.8, the estimate on the EZ dummy variable is more important than R^2 .

The R -squared from the dummy variable regression can be used to compute F tests in the usual way, assuming, of course, that the classical linear model assumptions hold (see the chapter appendix). In particular, we can test the joint significance of all of the cross-sectional dummies ($N - 1$, since one unit is chosen as the base group). The unrestricted R -squared is obtained from the regression with all of the cross-sectional dummies; the restricted R -squared omits these. In the vast majority of applications, the dummy variables will be jointly significant.

Occasionally, the estimated intercepts, say $\hat{a}_i$, are of interest. This is the case if we want to study the distribution of the $\hat{a}_i$ across i , or if we want to pick a particular firm or city to see whether its $\hat{a}_i$ is above or below the average value in the sample. These estimates are directly available from the dummy variable regression, but they are rarely reported by packages that have fixed effects routines (for the practical reason that there are so many $\hat{a}_i$). After fixed effects estimation with N of any size, the $\hat{a}_i$ are pretty easy to compute:

$$\hat{a}_i = \bar{y}_i - \hat{\beta}_1 \bar{x}_{i1} - \dots - \hat{\beta}_k \bar{x}_{ik}, i = 1, \dots, N, \quad [14.6]$$

where the overbar refers to the time averages and the $\hat{\beta}_j$ are the fixed effects estimates. For example, if we have estimated a model of crime while controlling for various time-varying factors, we can obtain $\hat{a}_i$ for a city to see whether the unobserved fixed effects that contribute to crime are above or below average.

Some econometrics packages that support fixed effects estimation report an “intercept,” which can cause confusion in light of our earlier claim that the time-demeaning eliminates all time-constant variables, including an overall intercept. [See equation (14.5).] Reporting an overall intercept in fixed effects (FE) estimation arises from viewing the a_i as parameters to estimate. Typically, the intercept reported is the average across i of the $\hat{a}_i$. In other words, the overall intercept is actually the average of the individual-specific intercepts, which is an unbiased, consistent estimator of $\alpha = E(a_i)$.

In most studies, the $\hat{\beta}_j$ are of interest, and so the time-demeaned equations are used to obtain these estimates. Further, it is usually best to view the a_i as omitted variables that we control for through the within transformation. The sense in which the a_i can be estimated is generally weak. In fact, even though $\hat{a}_i$ is unbiased (under Assumptions FE.1 through FE.4 in the chapter appendix), it is not consistent with a fixed T as $N \rightarrow \infty$. The reason is that, as we add each additional cross-sectional observation, we add a new a_i . No information accumulates on each a_i when T is fixed. With larger T , we can get better estimates of the a_i , but most panel data sets are of the large N and small T variety.

Fixed Effects or First Differencing?

So far, setting aside pooled OLS, we have seen two competing methods for estimating unobserved effects models. One involves differencing the data, and the other involves time-demeaning. How do we know which one to use?

We can eliminate one case immediately: when $T = 2$, the FE and FD estimates, as well as all test statistics, are *identical*, and so it does not matter which we use. Of course, the equivalence between the FE and FD estimates requires that we estimate the same model in each case. In particular, as we discussed in Chapter 13, it is natural to include an intercept in the FD equation; this intercept is actually the intercept for the second time period in the original model written for the two time periods. Therefore, FE estimation must include a dummy variable for the second time period in order to be identical to the FD estimates that include an intercept.

With $T = 2$, FD has the advantage of being straightforward to implement in any econometrics or statistical package that supports basic data manipulation, and it is easy to compute heteroskedasticity-robust statistics after FD estimation (because when $T = 2$, FD estimation is just a cross-sectional regression).

When $T \geq 3$, the FE and FD estimators are not the same. Since both are unbiased under Assumptions FE.1 through FE.4, we cannot use unbiasedness as a criterion. Further, both are consistent (with T fixed as $N \rightarrow \infty$) under FE.1 through FE.4. For large N and small T , the choice between FE and FD hinges on the relative efficiency of the estimators, and this is determined by the serial correlation in the idiosyncratic errors, u_{it} . (We will assume homoskedasticity of the u_{it} , since efficiency comparisons require homoskedastic errors.)

When the u_{it} are serially uncorrelated, fixed effects is more efficient than first differencing (and the standard errors reported from fixed effects are valid). Since the unobserved effects model is typically stated (sometimes only implicitly) with serially uncorrelated idiosyncratic errors, the FE estimator is used more than the FD estimator. But we should remember that this assumption can be false. In many applications, we can expect the unobserved factors that change over time to be serially correlated. If u_{it} follows a random walk—which means that there is very substantial, positive serial correlation—then the difference Δu_{it} is serially uncorrelated, and first differencing is better. In many cases, the u_{it} exhibit some positive serial correlation, but perhaps not as much as a random walk. Then, we cannot easily compare the efficiency of the FE and FD estimators.

It is difficult to test whether the u_{it} are serially uncorrelated after FE estimation: we can estimate the time-demeaned errors, $\tilde{u}_{it}$, but not the u_{it} . However, in Section 13.3, we showed how to test whether the differenced errors, Δu_{it} , are serially uncorrelated. If this seems to be the case, FD can be used. If there is substantial negative serial correlation in the Δu_{it} , FE is probably better. It is often a good idea to try both: if the results are not sensitive, so much the better.

When T is large, and especially when N is not very large (for example, $N = 20$ and $T = 30$), we must exercise caution in using the fixed effects estimator. Although exact distributional results hold for any N and T under the classical fixed effects assumptions, inference can be very sensitive to violations of the assumptions when N is small and T is large. In particular, if we are using unit root processes—see Chapter 11—the spurious regression problem can arise. First differencing has the advantage of turning an integrated time series process into a weakly dependent process. Therefore, if we apply first differencing, we can appeal to the central limit theorem even in cases where T is larger than N . Normality in the idiosyncratic errors is not needed, and heteroskedasticity and serial correlation can be dealt with as we touched on in Chapter 13. Inference with the fixed effects estimator is potentially more sensitive to nonnormality, heteroskedasticity, and serial correlation in the idiosyncratic errors.

Like the first difference estimator, the fixed effects estimator can be very sensitive to classical measurement error in one or more explanatory variables. However, if each x_{itj} is uncorrelated with u_{it} , but the strict exogeneity assumption is otherwise violated—for example, a lagged dependent variable is included among the regressors or there is feedback between u_{it} and future outcomes of the explanatory variable—then the FE estimator likely has substantially less bias than the FD estimator (unless $T = 2$). The important theoretical fact is that the bias in the FD estimator does not depend on T , while that for the FE estimator tends to zero at the rate $1/T$. See Wooldridge (2010, Section 10.7) for details.

Generally, it is difficult to choose between FE and FD when they give substantively different results. It makes sense to report both sets of results and to try to determine why they differ.

Fixed Effects with Unbalanced Panels

Some panel data sets, especially on individuals or firms, have missing years for at least some cross-sectional units in the sample. In this case, we call the data set an **unbalanced panel**. The mechanics of fixed effects estimation with an unbalanced panel are not much more difficult than with a balanced panel. If T_i is the number of time periods for cross-sectional unit i , we simply use these T_i observations in doing the time-demeaning. The total number of observations is then $T_1 + T_2 + \dots + T_N$. As in the balanced case, one degree of freedom is lost for every cross-sectional observation due to the time-demeaning. Any regression package that does fixed effects makes the appropriate adjustment for this loss. The dummy variable regression also goes through in exactly the same way as with a balanced panel, and the df is appropriately obtained.

It is easy to see that units for which we have only a single time period play no role in a fixed effects analysis. The time-demeaning for such observations yields all zeros, which are not used in the estimation. (If T_i is at most two for all i , we can use first differencing: if $T_i = 1$ for any i , we do not have two periods to difference.)

The more difficult issue with an unbalanced panel is determining why the panel is unbalanced. With cities and states, for example, data on key variables are sometimes missing for certain years. Provided the reason we have missing data for some i is not correlated with the idiosyncratic errors, u_{it} , the unbalanced panel causes no problems. When we have data on individuals, families, or firms, things are trickier. Imagine, for example, that we obtain a random sample of manufacturing firms in 1990, and we are interested in testing how unionization affects firm profitability. Ideally, we can use a panel data analysis to control for unobserved worker and management characteristics that affect profitability and might also be correlated with the fraction of the firm's work force that is unionized. If we collect data again in subsequent years, some firms may be lost because they have gone out of business or have merged with other companies. If so, we probably have a nonrandom sample in subsequent time periods. The question is: If we apply fixed effects to the unbalanced panel, when will the estimators be unbiased (or at least consistent)?

If the reason a firm leaves the sample (called *attrition*) is correlated with the idiosyncratic error—those unobserved factors that change over time and affect profits—then the resulting sample selection problem (see Chapter 9) can cause biased estimators. This is a serious consideration in this example. Nevertheless, one useful thing about a fixed effects analysis is that it *does* allow attrition to be correlated with a_i , the unobserved effect.

The idea is that, with the initial sampling, some units are more likely to drop out of the survey, and this is captured by a_i .

EXAMPLE 14.3 EFFECT OF JOB TRAINING ON FIRM SCRAP RATES

We add two variables to the analysis in Table 14.1: $\log(\text{sales}_{it})$ and $\log(\text{employ}_{it})$, where *sales* is annual firm sales and *employ* is number of employees. Three of the 54 firms drop out of the analysis entirely because they do not have sales or employment data. Five additional observations are lost due to missing data on one or both of these variables for some years, leaving us with $n = 148$. Using fixed effects on the unbalanced panel does not change the basic story, although the estimated grant effect gets larger: $\hat{\beta}_{\text{grant}} = -.297$, $t_{\text{grant}} = -1.89$; $\hat{\beta}_{\text{grant-1}} = .536$, $t_{\text{grant-1}} = -2.389$.

Solving general attrition problems in panel data is complicated and beyond the scope of this text. [See, for example, Wooldridge (2010, Chapter 19).]

14.2 Random Effects Models

We begin with the same unobserved effects model as before,

$$y_{it} = \beta_0 + \beta_1 x_{it1} + \dots + \beta_k x_{itk} + a_i + u_{it}, \quad [14.7]$$

where we explicitly include an intercept so that we can make the assumption that the unobserved effect, a_i , has zero mean (without loss of generality). We would usually allow for time dummies among the explanatory variables as well. In using fixed effects or first differencing, the goal is to eliminate a_i because it is thought to be correlated with one or more of the x_{itj} . But suppose we think a_i is *uncorrelated* with each explanatory variable in all time periods. Then, using a transformation to eliminate a_i results in inefficient estimators.

Equation (14.7) becomes a **random effects model** when we assume that the unobserved effect a_i is uncorrelated with each explanatory variable:

$$\text{Cov}(x_{itj}, a_i) = 0, \quad t = 1, 2, \dots, T; j = 1, 2, \dots, k. \quad [14.8]$$

In fact, the ideal random effects assumptions include all of the fixed effects assumptions plus the additional requirement that a_i is independent of all explanatory variables in all time periods. (See the chapter appendix for the actual assumptions used.) If we think the unobserved effect a_i is correlated with any explanatory variables, we should use first differencing or fixed effects.

Under (14.8) and along with the random effects assumptions, how should we estimate the β_j ? It is important to see that, if we believe that a_i is uncorrelated with the explanatory variables, the β_j can be consistently estimated by using a single cross section: there is no need for panel data at all. But using a single cross section disregards much useful information in the other time periods. We can also use the data in a pooled OLS procedure: just run OLS of y_{it} on the explanatory variables and probably the time dummies. This, too, produces consistent estimators of the β_j under the random effects assumption. But it ignores

a key feature of the model. If we define the **composite error term** as $v_{it} = a_i + u_{it}$, then (14.7) can be written as

$$y_{it} = \beta_0 + \beta_1 x_{it1} + \dots + \beta_k x_{itk} + v_{it} \quad [14.9]$$

Because a_i is in the composite error in each time period, the v_{it} are serially correlated across time. In fact, under the random effects assumptions,

$$\text{Corr}(v_{it}, v_{is}) = \sigma_a^2 / (\sigma_a^2 + \sigma_u^2), \quad t \neq s,$$

where $\sigma_a^2 = \text{Var}(a_i)$ and $\sigma_u^2 = \text{Var}(u_{it})$. This (necessarily) positive serial correlation in the error term can be substantial, and, because the usual pooled OLS standard errors ignore this correlation, they will be incorrect, as will the usual test statistics. In Chapter 12, we showed how generalized least squares can be used to estimate models with autoregressive serial correlation. We can also use GLS to solve the serial correlation problem here. For the procedure to have good properties, we should have large N and relatively small T . We assume that we have a balanced panel, although the method can be extended to unbalanced panels.

Deriving the GLS transformation that eliminates serial correlation in the errors requires sophisticated matrix algebra [see, for example, Wooldridge (2010, Chapter 10)]. But the transformation itself is simple. Define

$$\theta = 1 - [\sigma_u^2 / (\sigma_u^2 + T\sigma_a^2)]^{1/2}, \quad [14.10]$$

which is between zero and one. Then, the transformed equation turns out to be

$$\begin{aligned} y_{it} - \theta \bar{y}_i &= \beta_0(1 - \theta) + \beta_1(x_{it1} - \theta \bar{x}_{i1}) + \dots \\ &+ \beta_k(x_{itk} - \theta \bar{x}_{ik}) + (v_{it} - \theta \bar{v}_i), \end{aligned} \quad [14.11]$$

where the overbar again denotes the time averages. This is a very interesting equation, as it involves **quasi-demeaned data** on each variable. The fixed effects estimator subtracts the time averages from the corresponding variable. The random effects transformation subtracts a fraction of that time average, where the fraction depends on σ_u^2 , σ_a^2 , and the number of time periods, T . The GLS estimator is simply the pooled OLS estimator of equation (14.11). It is hardly obvious that the errors in (14.11) are serially uncorrelated, but they are. (See Problem 3.)

The transformation in (14.11) allows for explanatory variables that are constant over time, and this is one advantage of random effects (RE) over either fixed effects or first differencing. This is possible because RE assumes that the unobserved effect is uncorrelated with all explanatory variables, whether the explanatory variables are fixed over time or not. Thus, in a wage equation, we can include a variable such as education even if it does not change over time. But we are assuming that education is uncorrelated with a_i , which contains ability and family background. In many applications, the whole reason for using panel data is to allow the unobserved effect to be correlated with the explanatory variables.

The parameter θ is never known in practice, but it can always be estimated. There are different ways to do this, which may be based on pooled OLS or fixed effects, for example. Generally, $\hat{\theta}$ takes the form $\hat{\theta} = 1 - \{1/[1 + T(\sigma_a^2/\hat{\sigma}_u^2)]\}^{1/2}$, where $\hat{\sigma}_a^2$ is a consistent estimator of σ_a^2 and $\hat{\sigma}_u^2$ is a consistent estimator of σ_u^2 . These estimators can be based on the pooled OLS or fixed effects residuals. One possibility is that

$\hat{\sigma}_a^2 = [NT(T-1)/2 - (k+1)]^{-1} \sum_{i=1}^N \sum_{t=1}^{T-1} \sum_{s=t+1}^T \hat{v}_{it} \hat{v}_{is}$, where the $\hat{v}_{it}$ are the residuals from estimating (14.9) by pooled OLS. Given this, we can estimate σ_u^2 by using $\hat{\sigma}_v^2 = \hat{\sigma}_v^2 - \hat{\sigma}_a^2$, where $\hat{\sigma}_v^2$ is the square of the usual standard error of the regression from pooled OLS. [See Wooldridge (2010, Chapter 10) for additional discussion of these estimators.]

Many econometrics packages support estimation of random effects models and automatically compute some version of $\hat{\theta}$. The feasible GLS estimator that uses $\hat{\theta}$ in place of θ is called the **random effects estimator**. Under the random effects assumptions in the chapter appendix, the estimator is consistent (not unbiased) and asymptotically normally distributed as N gets large with fixed T . The properties of the random effects (RE) estimator with small N and large T are largely unknown, although it has certainly been used in such situations.

Equation (14.11) allows us to relate the RE estimator to both pooled OLS and fixed effects. Pooled OLS is obtained when $\theta = 0$, and FE is obtained when $\theta = 1$. In practice, the estimate $\hat{\theta}$ is never zero or one. But if $\hat{\theta}$ is close to zero, the RE estimates will be close to the pooled OLS estimates. This is the case when the unobserved effect, a_i , is relatively unimportant (because it has small variance relative to σ_u^2). It is more common for σ_a^2 to be large relative to σ_u^2 , in which case $\hat{\theta}$ will be closer to unity. As T gets large, $\hat{\theta}$ tends to one, and this makes the RE and FE estimates very similar.

We can gain more insight on the relative merits of random effects versus fixed effects by writing the quasi-demeaned error in equation (14.11) as $v_{it} - \theta \bar{v}_i = (1 - \theta)a_i + u_{it} - \theta \bar{u}_i$. This simple expression makes it clear that the errors in the transformed equation used in random effects estimation weight the unobserved effect by $(1 - \theta)$. Although correlation between a_i and one or more x_{ijt} causes inconsistency in the random effects estimation, we see that the correlation is attenuated by the factor $(1 - \theta)$. As $\theta \rightarrow 1$, the bias term goes to zero, as it must because the RE estimator tends to the FE estimator. If θ is close to zero, we are leaving a larger fraction of the unobserved effect in the error term, and, as a consequence, the asymptotic bias of the RE estimator will be larger.

In applications of FE and RE, it is usually informative also to compute the pooled OLS estimates. Comparing the three sets of estimates can help us determine the nature of the biases caused by leaving the unobserved effect, a_i , entirely in the error term (as does pooled OLS) or partially in the error term (as does the RE transformation). But we must remember that, even if a_i is uncorrelated with all explanatory variables in all time periods, the pooled OLS standard errors and test statistics are generally invalid: they ignore the often substantial serial correlation in the composite errors, $v_{it} = a_i + u_{it}$. As we mentioned in Chapter 13 (see Example 13.9), it is possible to compute standard errors and test statistics that are robust to arbitrary serial correlation (and heteroskedasticity) in v_{it} , and popular statistics packages often allow this option. [See, for example, Wooldridge (2010, Chapter 10).]

EXAMPLE 14.4

A WAGE EQUATION USING PANEL DATA

We again use the data in WAGEPAN.RAW to estimate a wage equation for men. We use three methods: pooled OLS, random effects, and fixed effects. In the first two methods, we can include *educ* and race dummies (*black* and *hispan*), but these drop out of the fixed effects analysis. The time-varying variables are *exper*, *exper*², *union*, and *married*. As we discussed in Section 14.1, *exper* is dropped in the FE analysis (although *exper*² remains). Each regression also contains a full set of year dummies. The estimation results are in Table 14.2.

TABLE 14.2 Three Different Estimators of a Wage Equation

Dependent Variable: $\log(\text{wage})$			
Independent Variables	Pooled OLS	Random Effects	Fixed Effects
<i>educ</i>	.091 (.005)	.092 (.011)	————
<i>black</i>	−.139 (.024)	−.139 (.048)	————
<i>hispan</i>	.016 (.021)	.022 (.043)	————
<i>exper</i>	.067 (.014)	.106 (.015)	————
<i>exper</i> ²	−.0024 (.0008)	−.0047 (.0007)	−.0052 (.0007)
<i>married</i>	.108 (.016)	.064 (.017)	.047 (.018)
<i>union</i>	.182 (.017)	.106 (.018)	.080 (.019)

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The coefficients on *educ*, *black*, and *hispan* are similar for the pooled OLS and random effects estimations. The pooled OLS standard errors are the usual OLS standard errors, and these underestimate the true standard errors because they ignore the positive serial correlation; we report them here for comparison only. The experience profile is somewhat different, and both the marriage and union premiums fall notably in the random effects estimation. When we eliminate the unobserved effect entirely by using fixed effects, the marriage premium falls to about 4.7%, although it is still statistically significant. The drop in the marriage premium is consistent with the idea that men who are more able—as captured by a higher unobserved effect, a_i —are more likely to be married. Therefore, in the pooled OLS estimation, a large part of the marriage premium reflects the fact that men who are married would earn more even if they were not married. The remaining 4.7% has at least two possible explanations: (1) marriage really makes men more productive or (2) employers pay married men a premium because marriage is a signal of stability. We cannot distinguish between these two hypotheses.

EXPLORING FURTHER 14.3

The union premium estimated by fixed effects is about 10 percentage points lower than the OLS estimate. What does this strongly suggest about the correlation between *union* and the unobserved effect?

The estimate of θ for the random effects estimation is $\hat{\theta} = .643$, which helps explain why, on the time-varying variables, the RE estimates lie closer to the FE estimates than to the pooled OLS estimates.

Random Effects or Fixed Effects?

Because fixed effects allows arbitrary correlation between a_i and the x_{itj} , while random effects does not, FE is widely thought to be a more convincing tool for estimating ceteris paribus effects. Still, random effects is applied in certain situations. Most obviously, if

the key explanatory variable is constant over time, we cannot use FE to estimate its effect on y . For example, in Table 14.2, we must rely on the RE (or pooled OLS) estimate of the return to education. Of course, we can only use random effects because we are willing to assume the unobserved effect is uncorrelated with all explanatory variables. Typically, if one uses random effects, as many time-constant controls as possible are included among the explanatory variables. (With an FE analysis, it is not necessary to include such controls.) RE is preferred to pooled OLS because RE is generally more efficient.

If our interest is in a time-varying explanatory variable, is there ever a case to use RE rather than FE? Yes, but situations in which $\text{Cov}(x_{ij}, a_i) = 0$ should be considered the exception rather than the rule. If the key policy variable is set experimentally—say, each year, children are randomly assigned to classes of different sizes—then random effects would be appropriate for estimating the effect of class size on performance. Unfortunately, in most cases the regressors are themselves outcomes of choice processes and likely to be correlated with individual preferences and abilities as captured by a_i .

It is still fairly common to see researchers apply both random effects and fixed effects, and then formally test for statistically significant differences in the coefficients on the time-varying explanatory variables. (So, in Table 14.2, these would be the coefficients on *exper*², *married*, and *union*.) Hausman (1978) first proposed such a test, and some econometrics packages routinely compute the Hausman test under the full set of random effects assumptions listed in the appendix to this chapter. The idea is that one uses the random effects estimates unless the Hausman test rejects (14.8). In practice, a failure to reject means either that the RE and FE estimates are sufficiently close so that it does not matter which is used, or the sampling variation is so large in the FE estimates that one cannot conclude practically significant differences are statistically significant. In the latter case, one is left to wonder whether there is enough information in the data to provide precise estimates of the coefficients. A rejection using the Hausman test is taken to mean that the key RE assumption, (14.8), is false, and then the FE estimates are used. (Naturally, as in all applications of statistical inference, one should distinguish between a practically significant difference and a statistically significant difference.) Wooldridge (2010, Chapter 10) contains further discussion. In the next section we discuss an alternative, computationally simpler approach to choosing between the RE and FE approaches.

A final word of caution. In reading empirical work, you may find that some authors decide on FE versus RE estimation based on whether the a_i are properly viewed as parameters to estimate or as random variables. Such considerations are usually wrongheaded. In this chapter, we have treated the a_i as random variables in the unobserved effects model (14.7), regardless of how we decide to estimate the β_j . As we have emphasized, the key issue that determines whether we use FE or RE is whether we can plausibly assume a_i is uncorrelated with all x_{ij} . Nevertheless, in some applications of panel data methods, we cannot treat our sample as a random sample from a large population, especially when the unit of observation is a large geographical unit (say, states or provinces). Then, it often makes sense to think of each a_i as a separate intercept to estimate for each cross-sectional unit. In this case, we use fixed effects: remember, using FE is mechanically the same as allowing a different intercept for each cross-sectional unit. Fortunately, whether or not we engage in the philosophical debate about the nature of a_i , FE is almost always much more convincing than RE for policy analysis using aggregated data.

14.3 The Correlated Random Effects Approach

In applications where it makes sense to view the a_i (unobserved effects) as being random variables, along with the observed variables we draw, there is an alternative to fixed effects that still allows a_i to be correlated with the observed explanatory variables. To describe the approach, consider again the simple model in equation (14.1), with a single, time-varying explanatory variable x_{it} . Rather than assume a_i is uncorrelated with $\{x_{it}; t = 1, 2, \dots, T\}$ —which is the random effects approach—or take away time averages to remove a_i —the fixed effects approach—we might instead model correlation between a_i and $\{x_{it}; t = 1, 2, \dots, T\}$. Because a_i is, by definition, constant over time, allowing it to be correlated with the average level of the x_{it} has a certain appeal. More specifically, let $\bar{x}_i = T^{-1}\sum_{t=1}^T x_{it}$ be the time average, as before. Suppose we assume the simple linear relationship

$$a_i = \alpha + \gamma\bar{x}_i + r_i \quad [14.12]$$

where we assume r_i is uncorrelated with each x_{it} . Because $\bar{x}_i$ is a linear function of the x_{it} ,

$$\text{Cov}(\bar{x}_i, r_i) = 0. \quad [14.13]$$

Equations (14.12) and (14.13) imply that a_i and $\bar{x}_i$ are correlated whenever $\gamma \neq 0$.

The **correlated random effects** (CRE) approach uses (14.12) in conjunction with (14.1): substituting the former in the latter gives

$$y_{it} = \beta x_{it} + \alpha + \gamma\bar{x}_i + r_i + u_{it} = \alpha + \beta x_{it} + \gamma\bar{x}_i + r_i + u_{it} \quad [14.14]$$

Equation (14.14) is interesting because it still has a composite error term, $r_i + u_{it}$, consisting of a time-constant unobservable r_i and the idiosyncratic shocks, u_{it} . Importantly, assumption (14.8) holds when we replace a_i with r_i . Also, because u_{it} is assumed to be uncorrelated with x_{is} , all s and t , u_{it} is also uncorrelated with $\bar{x}_i$. All of these assumptions add up to random effects estimation of the equation

$$y_{it} = \alpha + \beta x_{it} + \gamma\bar{x}_i + r_i + u_{it} \quad [14.15]$$

which is like the usual equation underlying RE estimation with the important addition of the time-average variable, $\bar{x}_i$. It is the addition of $\bar{x}_i$ that controls for the correlation between a_i and the sequence $\{x_{it}; t = 1, 2, \dots, T\}$. What is left over, r_i , is uncorrelated with the x_{it} .

In most econometrics packages it is easy to compute the unit-specific time averages, $\bar{x}_i$. Assuming we have done that for each cross-sectional unit i , what can we expect to happen if we apply RE to equation (14.15)? Notice that estimation of (14.15) gives $\hat{\alpha}_{CRE}$, $\hat{\beta}_{CRE}$, and $\hat{\gamma}_{CRE}$ —the CRE estimators. As far as $\hat{\beta}_{CRE}$ goes, the answer is a bit anticlimactic. It can be shown—see, for example, Wooldridge (2010, Chapter 10)—that

$$\hat{\beta}_{CRE} = \hat{\beta}_{FE} \quad [14.16]$$

where $\hat{\beta}_{FE}$ denotes the FE estimator from equation (14.3). In other words, adding the time average $\bar{x}_i$ and using random effects is the same as subtracting the time averages and using pooled OLS.

Even though (14.15) is not needed to obtain $\hat{\beta}_{FE}$, the equivalence of the CRE and FE estimates of β provides a nice interpretation of FE: it controls for the average level, $\bar{x}_i$, when measuring the partial effect of x_{it} on y_{it} . As an example, suppose that x_{it} is a tax rate on firm profits in county i in year t , and y_{it} is some measure of county-level economic output. By including $\bar{x}_i$, the average tax rate in the county over the T years, we are allowing for systematic differences between historically high-tax and low-tax counties—differences that may also affect economic output.

We can also use equation (14.15) to see why the FE estimators are often much less precise than the RE estimators. If we set $\gamma = 0$ in equation (14.15) then we obtain the usual RE estimator of β , $\hat{\beta}_{RE}$. This means that correlation between x_{it} and $\bar{x}_i$ has no bearing on the variance of the RE estimator. By contrast, we know from multiple regression analysis in Chapter 3 that correlation between x_{it} and $\bar{x}_i$ —that is, multicollinearity—can result in a higher variance for $\hat{\beta}_{FE}$. Sometimes the variance is much higher, particularly when there is little variation in x_{it} across t , in which case x_{it} and $\bar{x}_i$ tend to be highly correlated. In the limiting case where there is no variation across time for any i , the correlation is perfect—and FE fails to provide an estimate of β .

Apart from providing a synthesis of the FE and RE approaches, are there other reasons to consider the CRE approach even if it simply delivers the usual FE estimate of β ? Yes, at least two. First, the CRE approach provides a simple, formal way of choosing between the FE and RE approaches. As we just discussed, the RE approach sets $\gamma = 0$ while FE estimates γ . Because we have $\hat{\gamma}_{CRE}$ and its standard error [obtained from RE estimation of (14.15)], we can construct a t test of $H_0: \gamma = 0$ against $H_1: \gamma \neq 0$. [The appendix discusses how to make this test robust to heteroskedasticity and serial correlation in $\{u_{it}\}$.] If we reject H_0 at a sufficiently small significance level, we reject RE in favor of FE. As usual, especially with a large cross section, it is important to distinguish between a statistical rejection and economically important differences.

A second reason to study the CRE approach is that it provides a way to include time-constant explanatory variables in what is effectively a fixed effects analysis. For example, let z_i be a variable that does not change over time—it could be gender, say, or an IQ test score determined in childhood. We can easily augment (14.15) to include z_i :

$$y_{it} = \alpha + \beta x_{it} + \gamma \bar{x}_i + \delta z_i + r_i + u_{it}, \quad [14.17]$$

where we do not change the notation for the error term (which no longer includes z_i). If we estimate this expanded equation by RE, it can still be shown that the estimate of β is the FE estimate from (14.1). In fact, once we include $\bar{x}_i$, we can include any other time-constant variables in the equation, estimate it by RE, and obtain $\hat{\beta}_{FE}$ as the coefficient on x_{it} . In addition, we obtain an estimate of δ , although the estimate should be interpreted with caution because it does not necessarily estimate a causal effect of z_i on y_{it} .

The same CRE strategy can be applied to models with many time-varying explanatory variables (and many time-constant variables). When the equation augmented with the time averages is estimated by RE, the coefficients on the time-varying variables are identical to the FE estimates. As a practical note, when the panel is balanced there is no need to include the time averages of variables that change over time—the leading case being time period dummies. (With T time periods, the time average of a time period is just $1/T$, a constant for all i and t ; clearly it makes no sense to add a bunch of constants to an equation that already has an intercept.) If the panel data set is unbalanced, then the average of

variables such as time dummies can change across i —it will depend on how many periods we have for cross-sectional unit i . In such cases, the time averages of any variable that changes over time must be included.

Computer Exercise 14 in this chapter illustrates how the CRE approach can be applied to the balanced panel data set in AIRFARE.RAW, and how one can test RE versus FE in the CRE framework.

14.4 Applying Panel Data Methods to Other Data Structures

The various panel data methods can be applied to certain data structures that do not involve time. For example, it is common in demography to use siblings (sometimes twins) to account for unobserved family and background characteristics. Usually we want to allow the unobserved “family effect,” which is common to all siblings within a family, to be correlated with observed explanatory variables. If those explanatory variables vary across siblings within a family, differencing across sibling pairs—or, more generally, using the within transformation within a family—is preferred as an estimation method. By removing the unobserved effect, we eliminate potential bias caused by confounding family background characteristics. Implementing fixed effects on such data structures is rather straightforward in regression packages that support FE estimation.

As an example, Geronimus and Korenman (1992) used pairs of sisters to study the effects of teen childbearing on future economic outcomes. When the outcome is income relative to needs—something that depends on the number of children—the model is

$$\begin{aligned} \log(\text{incneeds}_{fs}) = & \beta_0 + \delta_0 \text{sister2}_s + \beta_1 \text{teenbrth}_{fs} \\ & + \beta_2 \text{age}_{fs} + \text{other factors} + a_f + u_{fs}, \end{aligned} \quad [14.18]$$

where f indexes family and s indexes a sister within the family. The intercept for the first sister is β_0 , and the intercept for the second sister is $\beta_0 + \delta_0$. The variable of interest is teenbrth_{fs} , which is a binary variable equal to one if sister s in family f had a child while a teenager. The variable age_{fs} is the current age of sister s in family f ; Geronimus and Korenman also used some other controls. The unobserved variable a_f , which changes only across family, is an *unobserved family effect* or a *family fixed effect*. The main concern in the analysis is that teenbrth is correlated with the family effect. If so, an OLS analysis that pools across families and sisters gives a biased estimator of the effect of teenage motherhood on economic outcomes. Solving this problem is simple: within each family, difference (14.18) across sisters to get

$$\Delta \log(\text{incneeds}) = \delta_0 + \beta_1 \Delta \text{teenbrth} + \beta_2 \Delta \text{age} + \dots + \Delta u; \quad [14.19]$$

this removes the family effect, a_f , and the resulting equation can be estimated by OLS. Notice that there is no time element here: the differencing is across sisters within a family. Also, we have allowed for differences in intercepts

EXPLORING FURTHER 14.4

When using the differencing method, does it make sense to include dummy variables for the mother and father’s race in (14.18)? Explain.

across sisters in (14.18), which leads to a nonzero intercept in the differenced equation, (14.19). If in entering the data the order of the sisters within each family is essentially random, the estimated intercept should be close to zero. But even in such cases it does not hurt to include an intercept in (14.19), and having the intercept allows for the fact that, say, the first sister listed might always be the neediest.

Using 129 sister pairs from the 1982 National Longitudinal Survey of Young Women, Geronimus and Korenman first estimated β_1 by pooled OLS to obtain $-.33$ or $-.26$, where the second estimate comes from controlling for family background variables (such as parents' education); both estimates are very statistically significant [see Table 3 in Geronimus and Korenman (1992)]. Therefore, teenage motherhood has a rather large impact on future family income. However, when the differenced equation is estimated, the coefficient on *teenbrth* is $-.08$, which is small and statistically insignificant. This suggests that it is largely a woman's family background that affects her future income, rather than teenage childbearing.

Geronimus and Korenman looked at several other outcomes and two other data sets; in some cases, the within family estimates were economically large and statistically significant. They also showed how the effects disappear entirely when the sisters' education levels are controlled for.

Ashenfelter and Krueger (1994) used the differencing methodology to estimate the return to education. They obtained a sample of 149 identical twins and collected information on earnings, education, and other variables. Identical twins were used because they should have the same underlying ability. This can be differenced away by using twin differences, rather than OLS on the pooled data. Because identical twins are the same in age, gender, and race, these factors all drop out of the differenced equation. Therefore, Ashenfelter and Krueger regressed the difference in $\log(\text{earnings})$ on the difference in education and estimated the return to education to be about 9.2% ($t = 3.83$). Interestingly, this is actually *larger* than the pooled OLS estimate of 8.4% (which controls for gender, age, and race). Ashenfelter and Krueger also estimated the equation by random effects and obtained 8.7% as the return to education. (See Table 5 in their paper.) The random effects analysis is mechanically the same as the panel data case with two time periods.

The samples used by Geronimus and Korenman (1992) and Ashenfelter and Krueger (1994) are examples of **matched pairs samples**. Generally, fixed and random effects methods can be applied to a **cluster sample**. These are cross-sectional data sets, but each observation belongs to a well-defined cluster. In the previous examples, each family is a cluster. As another example, suppose we have participation data on various pension plans, where firms offer more than one plan. We can then view each firm as a cluster, and it is pretty clear that unobserved firm effects would be an important factor in determining participation rates in pension plans within the firm.

Educational data on students sampled from many schools form a cluster sample, where each school is a cluster. Because the outcomes within a cluster are likely to be correlated, allowing for an unobserved cluster effect is typically important. Fixed effects estimation is preferred when we think the unobserved **cluster effect**—an example of which is a_f in (14.12)—is correlated with one or more of the explanatory variables. Then, we can only include explanatory variables that vary, at least somewhat, within clusters. The cluster sizes are rarely the same, so fixed effects methods for unbalanced panels are usually required.

In some cases, the key explanatory variables—often policy variables—change only at the level of the cluster, not within the cluster. In such cases the fixed effects approach is not applicable. For example, we may be interested in the effects of measured teacher quality on student performance, where each cluster is an elementary school classroom. Because

all students within a cluster have the same teacher, eliminating a “class effect” also eliminates any observed measures of teacher quality. If we have good controls in the equation, we may be justified in applying random effects on the unbalanced cluster. As with panel data, the key requirement for RE to produce convincing estimates is that the explanatory variables are uncorrelated with the unobserved cluster effect. Most econometrics packages allow random effects estimation on unbalanced clusters without much effort.

Pooled OLS is also commonly applied to cluster samples when eliminating a cluster effect via fixed effects is infeasible or undesirable. However, as with panel data, the usual OLS standard errors are incorrect unless there is no cluster effect, and so robust standard errors that allow “cluster correlation” (and heteroskedasticity) should be used. Some regression packages have simple commands to correct standard errors and the usual test statistics for general within cluster correlation (as well as heteroskedasticity). These are the same corrections that work for pooled OLS on panel data sets, which we reported in Example 13.9. As an example, Papke (1999) estimates linear probability models for the continuation of defined benefit pension plans based on whether firms adopted defined contribution plans. Because there is likely to be a firm effect that induces correlation across different plans within the same firm, Papke corrects the usual OLS standard errors for cluster sampling, as well as for heteroskedasticity in the linear probability model.

Summary

In this chapter we have continued our discussion of panel data methods, studying the fixed effects and random effects estimators, and also described the correlated random effects approach as a unifying framework. Compared with first differencing, the fixed effects estimator is efficient when the idiosyncratic errors are serially uncorrelated (as well as homoskedastic), and we make no assumptions about correlation between the unobserved effect a_i and the explanatory variables. As with first differencing, any time-constant explanatory variables drop out of the analysis. Fixed effects methods apply immediately to unbalanced panels, but we must assume that the reasons some time periods are missing are not systematically related to the idiosyncratic errors.

The random effects estimator is appropriate when the unobserved effect is thought to be uncorrelated with all the explanatory variables. Then, a_i can be left in the error term, and the resulting serial correlation over time can be handled by generalized least squares estimation. Conveniently, feasible GLS can be obtained by a pooled regression on quasi-demeaned data. The value of the estimated transformation parameter, $\hat{\theta}$, indicates whether the estimates are likely to be closer to the pooled OLS or the fixed effects estimates. If the full set of random effects assumptions holds, the random effects estimator is asymptotically—as N gets large with T fixed—more efficient than pooled OLS, first differencing, or fixed effects (which are all unbiased, consistent, and asymptotically normal).

The correlated random effects approach to panel data models has become more popular in recent years, primarily because it allows a simple test for choosing between FE and RE, and it allows one to incorporate time-constant variables in an equation that delivers the FE estimates of the time-varying variables. Finally, the panel data methods studied in Chapters 13 and 14 can be used when working with matched pairs or cluster samples. Differencing or the within transformation eliminates the cluster effect. If the cluster effect is uncorrelated with the explanatory variables, pooled OLS can be used, but the standard errors and test statistics should be adjusted for cluster correlation. Random effects estimation is also a possibility.

Key Terms

Cluster Effect	Fixed Effects Transformation	Unbalanced Panel
Cluster Sample	Matched Pairs Samples	Unobserved Effects Model
Composite Error Term	Quasi-Demeaned Data	Within Estimator
Correlated Random Effects	Random Effects Estimator	Within Transformation
Dummy Variable Regression	Random Effects Model	
Fixed Effects Estimator	Time-Demeaned Data	

Problems

- 1 Suppose that the idiosyncratic errors in (14.4), $\{u_{it}; t = 1, 2, \dots, T\}$, are serially uncorrelated with constant variance, σ_u^2 . Show that the correlation between adjacent differences, Δu_{it} and $\Delta u_{i,t+1}$, is $-.5$. Therefore, under the ideal FE assumptions, first differencing induces negative serial correlation of a known value.

- 2 With a single explanatory variable, the equation used to obtain the between estimator is

$$\bar{y}_i = \beta_0 + \beta_1 \bar{x}_i + a_i + \bar{u}_i,$$

where the overbar represents the average over time. We can assume that $E(a_i) = 0$ because we have included an intercept in the equation. Suppose that $\bar{u}_i$ is uncorrelated with $\bar{x}_i$, but $\text{Cov}(x_{it}, a_i) = \sigma_{xa}$ for all t (and i because of random sampling in the cross section).

- (i) Letting $\tilde{\beta}_1$ be the between estimator, that is, the OLS estimator using the time averages, show that

$$\text{plim } \tilde{\beta}_1 = \beta_1 + \sigma_{xa}/\text{Var}(\bar{x}_i),$$

where the probability limit is defined as $N \rightarrow \infty$. [Hint: See equations (5.5) and (5.6).]

- (ii) Assume further that the x_{it} , for all $t = 1, 2, \dots, T$, are uncorrelated with constant variance σ_x^2 . Show that $\text{plim } \tilde{\beta}_1 = \beta_1 + T(\sigma_{xa}/\sigma_x^2)$.
- (iii) If the explanatory variables are not very highly correlated across time, what does part (ii) suggest about whether the inconsistency in the between estimator is smaller when there are more time periods?
- 3 In a random effects model, define the composite error $v_{it} = a_i + u_{it}$, where a_i is uncorrelated with u_{it} and the u_{it} have constant variance σ_u^2 and are serially uncorrelated. Define $e_{it} = v_{it} - \theta \bar{v}_i$, where θ is given in (14.10).
- (i) Show that $E(e_{it}) = 0$.
- (ii) Show that $\text{Var}(e_{it}) = \sigma_u^2$, $t = 1, \dots, T$.
- (iii) Show that for $t \neq s$, $\text{Cov}(e_{it}, e_{is}) = 0$.
- 4 In order to determine the effects of collegiate athletic performance on applicants, you collect data on applications for a sample of Division I colleges for 1985, 1990, and 1995.
- (i) What measures of athletic success would you include in an equation? What are some of the timing issues?
- (ii) What other factors might you control for in the equation?

- (iii) Write an equation that allows you to estimate the effects of athletic success on the percentage change in applications. How would you estimate this equation? Why would you choose this method?
- 5 Suppose that, for one semester, you can collect the following data on a random sample of college juniors and seniors for each class taken: a standardized final exam score, percentage of lectures attended, a dummy variable indicating whether the class is within the student's major, cumulative grade point average prior to the start of the semester, and SAT score.
- Why would you classify this data set as a cluster sample? Roughly, how many observations would you expect for the typical student?
 - Write a model, similar to equation (14.18), that explains final exam performance in terms of attendance and the other characteristics. Use s to subscript student and c to subscript class. Which variables do not change within a student?
 - If you pool all of the data and use OLS, what are you assuming about unobserved student characteristics that affect performance and attendance rate? What roles do SAT score and prior GPA play in this regard?
 - If you think SAT score and prior GPA do not adequately capture student ability, how would you estimate the effect of attendance on final exam performance?
- 6 Using the “cluster” option in the econometrics package Stata[®] 11, the fully robust standard errors for the pooled OLS estimates in Table 14.2—that is, robust to serial correlation and heteroskedasticity in the composite errors, $\{v_{it}: t = 1, \dots, T\}$ —are obtained as $\text{se}(\hat{\beta}_{educ}) = .011$, $\text{se}(\hat{\beta}_{black}) = .051$, $\text{se}(\hat{\beta}_{hispan}) = .039$, $\text{se}(\hat{\beta}_{exper}) = .020$, $\text{se}(\hat{\beta}_{exper2}) = .0010$, $\text{se}(\hat{\beta}_{married}) = .026$, and $\text{se}(\hat{\beta}_{union}) = .027$.
- How do these standard errors generally compare with the nonrobust ones, and why?
 - How do the robust standard errors for pooled OLS compare with the standard errors for RE? Does it seem to matter whether the explanatory variable is time-constant or time-varying?
 - When the fully robust standard errors for the RE estimates are computed, Stata[®] 11 reports the following (where we look at only the coefficients on the time-varying variables): $\text{se}(\hat{\beta}_{exper}) = 0.16$, $\text{se}(\hat{\beta}_{expersq}) = .0008$, $\text{se}(\hat{\beta}_{married}) = 0.19$, and $\text{se}(\hat{\beta}_{union}) = 0.21$. [These are robust to any kind of serial correlation or heteroskedasticity in the idiosyncratic errors $\{u_{it}: t = 1, \dots, T\}$ as well as heteroskedasticity in a_i .] How do the robust standard errors generally compare with the usual RE standard errors reported in Table 2? What conclusion might you draw?
 - Comparing the four standard errors in part (iii) with their pooled OLS counterparts, what do you make of the fact that the robust RE standard errors are all below the robust POLS standard errors?

Computer Exercises

- C1 Use the data in RENTAL.RAW for this exercise. The data on rental prices and other variables for college towns are for the years 1980 and 1990. The idea is to see whether a stronger presence of students affects rental rates. The unobserved effects model is

$$\begin{aligned}\log(\text{rent}_{it}) = & \beta_0 + \delta_0 y90_t + \beta_1 \log(\text{pop}_{it}) + \beta_2 \log(\text{avginc}_i) \\ & + \beta_3 \text{pctstu}_{it} + a_i + u_{it},\end{aligned}$$

where *pop* is city population, *avginc* is average income, and *pctstu* is student population as a percentage of city population (during the school year).

- (i) Estimate the equation by pooled OLS and report the results in standard form. What do you make of the estimate on the 1990 dummy variable? What do you get for $\hat{\beta}_{pctstu}$?
- (ii) Are the standard errors you report in part (i) valid? Explain.
- (iii) Now, difference the equation and estimate by OLS. Compare your estimate of β_{pctstu} with that from part (i). Does the relative size of the student population appear to affect rental prices?
- (iv) Estimate the model by fixed effects to verify that you get identical estimates and standard errors to those in part (iii).

C2 Use CRIME4.RAW for this exercise.

- (i) Reestimate the unobserved effects model for crime in Example 13.9 but use fixed effects rather than differencing. Are there any notable sign or magnitude changes in the coefficients? What about statistical significance?
- (ii) Add the logs of each wage variable in the data set and estimate the model by fixed effects. How does including these variables affect the coefficients on the criminal justice variables in part (i)?
- (iii) Do the wage variables in part (ii) all have the expected sign? Explain. Are they jointly significant?

C3 For this exercise, we use JTRAIN.RAW to determine the effect of the job training grant on hours of job training per employee. The basic model for the three years is

$$hrsemp_{it} = \beta_0 + \delta_1 d88_t + \delta_2 d89_t + \beta_1 grant_{it} + \beta_2 grant_{it-1} + \beta_3 \log(employ_{it}) + a_i + u_{it}.$$

- (i) Estimate the equation using fixed effects. How many firms are used in the FE estimation? How many total observations would be used if each firm had data on all variables (in particular, *hrsemp*) for all three years?
- (ii) Interpret the coefficient on *grant* and comment on its significance.
- (iii) Is it surprising that *grant*₋₁ is insignificant? Explain.
- (iv) Do larger firms provide their employees with more or less training, on average? How big are the differences? (For example, if a firm has 10% more employees, what is the change in average hours of training?)

C4 In Example 13.8, we used the unemployment claims data from Papke (1994) to estimate the effect of enterprise zones on unemployment claims. Papke also uses a model that allows each city to have its own time trend:

$$\log(uclms_{it}) = a_i + c_i t + \beta_1 ez_{it} + u_{it},$$

where a_i and c_i are both unobserved effects. This allows for more heterogeneity across cities.

- (i) Show that, when the previous equation is first differenced, we obtain

$$\Delta \log(uclms_{it}) = c_i + \beta_1 \Delta ez_{it} + \Delta u_{it}, \quad t = 2, \dots, T.$$

Notice that the differenced equation contains a fixed effect, c_i .

- (ii) Estimate the differenced equation by fixed effects. What is the estimate of β_1 ? Is it very different from the estimate obtained in Example 13.8? Is the effect of enterprise zones still statistically significant?

- (iii) Add a full set of year dummies to the estimation in part (ii). What happens to the estimate of β_1 ?
- C5** (i) In the wage equation in Example 14.4, explain why dummy variables for occupation might be important omitted variables for estimating the union wage premium.
- (ii) If every man in the sample stayed in the same occupation from 1981 through 1987, would you need to include the occupation dummies in a fixed effects estimation? Explain.
- (iii) Using the data in WAGEPAN.RAW, include eight of the occupation dummy variables in the equation and estimate the equation using fixed effects. Does the coefficient on *union* change by much? What about its statistical significance?
- C6** Add the interaction term $union_{it} \cdot t$ to the equation estimated in Table 14.2 to see if wage growth depends on union status. Estimate the equation by random and fixed effects and compare the results.
- C7** Use the state-level data on murder rates and executions in MURDER.RAW for the following exercise.
- (i) Consider the unobserved effects model

$$mrd rte_{it} = \eta_t + \beta_1 exec_{it} + \beta_2 unem_{it} + a_i + u_{it},$$

- where η_t simply denotes different year intercepts and a_i is the unobserved state effect. If past executions of convicted murderers have a deterrent effect, what should be the sign of β_1 ? What sign do you think β_2 should have? Explain.
- (ii) Using just the years 1990 and 1993, estimate the equation from part (i) by pooled OLS. Ignore the serial correlation problem in the composite errors. Do you find any evidence for a deterrent effect?
- (iii) Now, using 1990 and 1993, estimate the equation by fixed effects. You may use first differencing since you are only using two years of data. Is there evidence of a deterrent effect? How strong?
- (iv) Compute the heteroskedasticity-robust standard error for the estimation in part (ii).
- (v) Find the state that has the largest number for the execution variable in 1993. (The variable *exec* is total executions in 1991, 1992, and 1993.) How much bigger is this value than the next highest value?
- (vi) Estimate the equation using first differencing, dropping Texas from the analysis. Compute the usual and heteroskedasticity-robust standard errors. Now, what do you find? What is going on?
- (vii) Use all three years of data and estimate the model by fixed effects. Include Texas in the analysis. Discuss the size and statistical significance of the deterrent effect compared with only using 1990 and 1993.
- C8** Use the data in MATHPNL.RAW for this exercise. You will do a fixed effects version of the first differencing done in Computer Exercises 11 in Chapter 13. The model of interest is

$$math4_{it} = \delta_1 y94_t + \dots + \delta_5 y98_t + \gamma_1 \log(rexpp_{it}) + \gamma_2 \log(rexpp_{it-1}) \\ + \psi_1 \log(enrol_{it}) + \psi_2 lunch_{it} + a_i + u_{it},$$

where the first available year (the base year) is 1993 because of the lagged spending variable.

- (i) Estimate the model by pooled OLS and report the usual standard errors. You should include an intercept along with the year dummies to allow a_i to have a nonzero expected value. What are the estimated effects of the spending variables? Obtain the OLS residuals, $\hat{v}_{it}$.
- (ii) Is the sign of the $lunch_{it}$ coefficient what you expected? Interpret the magnitude of the coefficient. Would you say that the district poverty rate has a big effect on test pass rates?
- (iii) Compute a test for AR(1) serial correlation using the regression $\hat{v}_{it}$ on $\hat{v}_{i,t-1}$. You should use the years 1994 through 1998 in the regression. Verify that there is strong positive serial correlation and discuss why.
- (iv) Now, estimate the equation by fixed effects. Is the lagged spending variable still significant?
- (v) Why do you think, in the fixed effects estimation, the enrollment and lunch program variables are jointly insignificant?
- (vi) Define the total, or long-run, effect of spending as $\theta_1 = \gamma_1 + \gamma_2$. Use the substitution $\gamma_1 = \theta_1 - \gamma_2$ to obtain a standard error for θ_1 . [Hint: Standard fixed effects estimation using $\log(rexpp_{it})$ and $z_{it} = \log(rexpp_{i,t-1}) - \log(rexpp_{it})$ as explanatory variables should do it.]

C9 The file PENSION.RAW contains information on participant-directed pension plans for U.S. workers. Some of the observations are for couples within the same family, so this data set constitutes a small cluster sample (with cluster sizes of two).

- (i) Ignoring the clustering by family, use OLS to estimate the model

$$\begin{aligned} pctstck = & \beta_0 + \beta_1 choice + \beta_2 prftshr + \beta_3 female + \beta_4 age \\ & + \beta_5 educ + \beta_6 finc25 + \beta_7 finc35 + \beta_8 finc50 + \beta_9 finc75 \\ & + \beta_{10} finc100 + \beta_{11} finc101 + \beta_{12} wealth89 + \beta_{13} stckin89 \\ & + \beta_{14} irain89 + u, \end{aligned}$$

where the variables are defined in the data set. The variable of most interest is *choice*, which is a dummy variable equal to one if the worker has a choice in how to allocate pension funds among different investments. What is the estimated effect of *choice*? Is it statistically significant?

- (ii) Are the income, wealth, stock holding, and IRA holding control variables important? Explain.
- (iii) Determine how many different families there are in the data set.
- (iv) Now, obtain the standard errors for OLS that are robust to cluster correlation within a family. Do they differ much from the usual OLS standard errors? Are you surprised?
- (v) Estimate the equation by differencing across only the spouses within a family. Why do the explanatory variables asked about in part (ii) drop out in the first-differenced estimation?
- (vi) Are any of the remaining explanatory variables in part (v) significant? Are you surprised?

C10 Use the data in AIRFARE.RAW for this exercise. We are interested in estimating the model

$$\begin{aligned} \log(fare_{it}) = & \eta_t + \beta_1 concen_{it} + \beta_2 \log(dist_i) + \beta_3 [\log(dist_i)]^2 \\ & + a_i + u_{it}, t = 1, \dots, 4, \end{aligned}$$

where η_t means that we allow for different year intercepts.

- (i) Estimate the above equation by pooled OLS, being sure to include year dummies. If $\Delta \text{concen} = .10$, what is the estimated percentage increase in *fare*?
- (ii) What is the usual OLS 95% confidence interval for β_1 ? Why is it probably not reliable? If you have access to a statistical package that computes fully robust standard errors, find the fully robust 95% CI for β_1 . Compare it to the usual CI and comment.
- (iii) Describe what is happening with the quadratic in $\log(\text{dist})$. In particular, for what value of *dist* does the relationship between $\log(\text{fare})$ and *dist* become positive? [Hint: First figure out the turning point value for $\log(\text{dist})$, and then exponentiate.] Is the turning point outside the range of the data?
- (iv) Now estimate the equation using random effects. How does the estimate of β_1 change?
- (v) Now estimate the equation using fixed effects. What is the FE estimate of β_1 ? Why is it fairly similar to the RE estimate? (Hint: What is $\hat{\theta}$ for RE estimation?)
- (vi) Name two characteristics of a route (other than distance between stops) that are captured by a_i . Might these be correlated with concen_i ?
- (vii) Are you convinced that higher concentration on a route increases airfares? What is your best estimate?

C11 This question assumes that you have access to a statistical package that computes standard errors robust to arbitrary serial correlation and heteroskedasticity for panel data methods.

- (i) For the pooled OLS estimates in Table 14.1, obtain the standard errors that allow for arbitrary serial correlation (in the composite errors, $v_{it} = a_i + u_{it}$) and heteroskedasticity. How do the robust standard errors for *educ*, *married*, and *union* compare with the nonrobust ones?
- (ii) Now obtain the robust standard errors for the fixed effects estimates that allow arbitrary serial correlation and heteroskedasticity in the idiosyncratic errors, u_{it} . How do these compare with the nonrobust FE standard errors?
- (iii) For which method, pooled OLS or FE, is adjusting the standard errors for serial correlation more important? Why?

C12 Use the data in ELEM94_95 to answer this question. The data are on elementary schools in Michigan. In this exercise, we view the data as a cluster sample, where each school is part of a district cluster.

- (i) What are the smallest and largest number of schools in a district? What is the average number of schools per district?
- (ii) Using pooled OLS (that is, pooling across all 1,848 schools), estimate a model relating *lavgsal* to *bs*, *lenrol*, *lstaff*, and *lunch*; see also Computer Exercises 11 from Chapter 9. What are the coefficient and standard error on *bs*?
- (iii) Obtain the standard errors that are robust to cluster correlation within district (and also heteroskedasticity). What happens to the *t* statistic for *bs*?
- (iv) Still using pooled OLS, drop the four observations with *bs* > .5 and obtain $\hat{\beta}_{bs}$ and its cluster-robust standard error. Now is there much evidence for a salary-benefits tradeoff?
- (v) Estimate the equation by fixed effects, allowing for a common district effect for schools within a district. Again drop the observations with *bs* > .5. Now what do you conclude about the salary-benefits tradeoff?
- (vi) In light of your estimates from parts (iv) and (v), discuss the importance of allowing teacher compensation to vary systematically across districts via a district fixed effect.

C13 The data set `DRIVING.RAW` includes state-level panel data (for the 48 continental U.S. states) from 1980 through 2004, for a total of 25 years. Various driving laws are indicated in the data set, including the alcohol level at which drivers are considered legally intoxicated. There are also indicators for “per se” laws—where licenses can be revoked without a trial—and seat belt laws. Some economics and demographic variables are also included.

- (i) How is the variable *totfatrte* defined? What is the average of this variable in the years 1980, 1992, and 2004? Run a regression of *totfatrte* on dummy variables for the years 1981 through 2004, and describe what you find. Did driving become safer over this period? Explain.
- (ii) Add the variables *bac08*, *bac10*, *perse*, *sbprim*, *sbsecon*, *sl70plus*, *gdl*, *perc14_24*, *unem*, and *vehicmilespc* to the regression from part (i). Interpret the coefficients on *bac8* and *bac10*. Do per se laws have a negative effect on the fatality rate? What about having a primary seat belt law? (Note that if a law was enacted sometime within a year the fraction of the year is recorded in place of the zero-one indicator.)
- (iii) Reestimate the model from part (ii) using fixed effects (at the state level). How do the coefficients on *bac08*, *bac10*, *perse*, and *sbprim* compare with the pooled OLS estimates? Which set of estimates do you think is more reliable?
- (iv) Suppose that *vehicmilespc*, the number of miles driven per capita, increases by 1,000. Using the FE estimates, what is the estimated effect on *totfatrte*? Be sure to interpret the estimate as if explaining to a layperson.
- (v) If there is serial correlation or heteroskedasticity in the idiosyncratic errors of the model then the standard errors in part (iii) are invalid. If possible, use “cluster” robust standard errors for the fixed effects estimates. What happens to the statistical significance of the policy variables in part (iii)?

C14 Use the data set in `AIRFARE.RAW` to answer this question. The estimates can be compared with those in Computer Exercise 10, in this Chapter.

- (i) Compute the time averages of the variable *concen*; call these *concenbar*. How many different time averages can there be? Report the smallest and the largest.
- (ii) Estimate the equation

$$lfare_{it} = \beta_0 + \delta_{1y98t} + \delta_{2y99t} + \delta_{3y00t} + \beta_1 concen_{it} + \beta_2 ldist_i + \beta_3 ldistsq_i + \gamma_1 concenbar_i + a_i + u_{it}$$
 by random effects. Verify that $\hat{\beta}_1$ is identical to the FE estimate computed in C10.
- (iii) If you drop *ldist* and *ldistsq* from the estimation in part (i) but still include *concenbar_i*, what happens to the estimate of $\hat{\beta}_1$? What happens to the estimate of γ_1 ?
- (iv) Using the equation in part (ii) and the usual RE standard error, test $H_0: \gamma_1 = 0$ against the two-sided alternative. Report the *p*-value. What do you conclude about RE versus FE for estimating β_1 in this application?
- (v) If possible, for the test in part (iv) obtain a *t*-statistic (and, therefore, *p*-value) that is robust to arbitrary serial correlation and heteroskedasticity. Does this change the conclusion reached in part (iv)?

APPENDIX 14A

14A.1 Assumptions for Fixed and Random Effects

In this appendix, we provide statements of the assumptions for fixed and random effects estimation. We also provide a discussion of the properties of the estimators under different sets of assumptions. Verification of these claims is somewhat involved, but can be found in Wooldridge (2010, Chapter 10).

Assumption FE.1

For each i , the model is

$$y_{it} = \beta_1 x_{it1} + \dots + \beta_k x_{itk} + a_i + u_{it}, \quad t = 1, \dots, T,$$

where the β_j are the parameters to estimate and a_i is the unobserved effect.

Assumption FE.2

We have a random sample from the cross section.

Assumption FE.3

Each explanatory variable changes over time (for at least some i), and no perfect linear relationships exist among the explanatory variables.

Assumption FE.4

For each t , the expected value of the idiosyncratic error given the explanatory variables in *all* time periods and the unobserved effect is zero: $E(u_{it} | \mathbf{X}_i, a_i) = 0$.

Under these first four assumptions—which are identical to the assumptions for the first-differencing estimator—the fixed effects estimator is unbiased. Again, the key is the strict exogeneity assumption, FE.4. Under these same assumptions, the FE estimator is consistent with a fixed T as $N \rightarrow \infty$.

Assumption FE.5

$\text{Var}(u_{it} | \mathbf{X}_i, a_i) = \text{Var}(u_{it}) = \sigma_u^2$, for all $t = 1, \dots, T$.

Assumption FE.6

For all $t \neq s$, the idiosyncratic errors are uncorrelated (conditional on all explanatory variables and a_i): $\text{Cov}(u_{it}, u_{is} | \mathbf{X}_i, a_i) = 0$.

Under Assumptions FE.1 through FE.6, the fixed effects estimator of the β_j is the best linear unbiased estimator. Since the FD estimator is linear and unbiased, it is necessarily worse than the FE estimator. The assumption that makes FE better than FD is FE.6, which implies that the idiosyncratic errors are serially uncorrelated.

Assumption FE.7

Conditional on $\mathbf{X}_i$ and a_i , the u_{it} are independent and identically distributed as $\text{Normal}(0, \sigma_u^2)$.

Assumption FE.7 implies FE.4, FE.5, and FE.6, but it is stronger because it assumes a normal distribution for the idiosyncratic errors. If we add FE.7, the FE estimator is normally distributed, and t and F statistics have exact t and F distributions. Without FE.7, we can rely on asymptotic approximations. But, without making special assumptions, these approximations require large N and small T .

The ideal random effects assumptions include FE.1, FE.2, FE.4, FE.5, and FE.6. (FE.7 could be added but it gains us little in practice because we have to estimate θ .) Because we are only subtracting a fraction of the time averages, we can now allow time-constant explanatory variables. So, FE.3 is replaced with

Assumption RE.1

There are no perfect linear relationships among the explanatory variables.

The cost of allowing time-constant regressors is that we must add assumptions about how the unobserved effect, a_i , is related to the explanatory variables.

Assumption RE.2

In addition to FE.4, the expected value of a_i given all explanatory variables is constant: $E(a_i | \mathbf{X}_i) = \beta_0$.

This is the assumption that rules out correlation between the unobserved effect and the explanatory variables, and it is the key distinction between fixed effects and random effects. Because we are assuming a_i is uncorrelated with all elements of $\mathbf{x}_{it}$, we can include time-constant explanatory variables. (Technically, the quasi-time-demeaning only removes a fraction of the time average, and not the whole time average.) We allow for a nonzero expectation for a_i in stating Assumption RE.4 so that the model under the random effects assumptions contains an intercept, β_0 , as in equation (14.7). Remember, we would typically include a set of time-period intercepts, too, with the first year acting as the base year.

We also need to impose homoskedasticity on a_i as follows:

Assumption RE.3

In addition to FE.5, the variance of a_i given all explanatory variables is constant:

$$\text{Var}(a_i | \mathbf{X}_i) = \sigma_a^2.$$

Under the six random effects assumptions (FE.1, FE.2, RE.3, RE.4, RE.5, and FE.6), the RE estimator is consistent and asymptotically normally distributed as N gets large for fixed T . Actually, consistency and asymptotic normality follow under the first four assumptions, but without the last two assumptions the usual RE standard errors and test statistics would not be valid. In addition, under the six RE assumptions, the RE estimators are asymptotically efficient. This means that, in large samples, the RE estimators will have smaller standard errors than the corresponding pooled OLS estimators (when the proper, robust standard errors are used for pooled OLS). For coefficients on time-varying explanatory variables (the only ones estimable by FE), the RE estimator is more efficient

than the FE estimator—often much more efficient. But FE is not meant to be efficient under the RE assumptions; FE is intended to be robust to correlation between a_i and the x_{it} . As often happens in econometrics, there is a tradeoff between robustness and efficiency. See Wooldridge (2010, Chapter 10) for verification of the claims made here.

14A.2 Inference Robust to Serial Correlation and Heteroskedasticity for Fixed Effects and Random Effects

One of the key assumptions for performing inference using the FE, RE, and even the CRE approach to panel data models is the assumption of no serial correlation in the idiosyncratic errors, $\{u_{it}; t = 1, \dots, T\}$ —see Assumption FE.6. Of course, heteroskedasticity can also be an issue, but this is also ruled out for standard inference (see Assumption FE.5). As discussed in the appendix to Chapter 13, the same issues can arise with first differencing estimation when we have $T \geq 3$ time periods.

Fortunately, as with FD estimation, there are now simple solutions for fully robust inference—inference that is robust to arbitrary violations of Assumptions FE.5 and FE.6 and, when applying the RE or CRE approaches, to Assumption RE.5. As with FD estimation, the general approach to obtaining fully robust standard errors and test statistics is known as **clustering**. Now, however, the clustering is applied to a different equation. For example, for FE estimation, the clustering is applied to the time-demeaned equation (14.5). For RE estimation, the clustering gets applied to the quasi-time-demeaned equation (14.11) [and a similar comment holds for CRE, but there the time averages are included as separate explanatory variables]. The details, which can be found in Wooldridge (2010, Chapter 10) are too advanced for this course. But understanding the purpose of clustering is not: if possible, we should compute standard errors, confidence intervals, and test statistics that are valid in large cross sections under the weakest set of assumptions. The FE estimator requires only Assumptions FE.1 to FE.4 for unbiasedness and consistency (as $N \rightarrow \infty$ with T fixed). Thus, a careful researcher at least checks whether inference made robust to serial correlation and heteroskedasticity in the errors affects inference. Experience shows that it often does.

Applying cluster robust inference to account for serial correlation within a panel data context is easily justified when N is substantially larger than T , but cannot be justified when N is small and T is larger. Computing the cluster robust statistics after FE or RE estimation is simple in many econometrics packages, often only requiring a qualification for the form “cluster(id)” appended to the end of FE and RE estimation commands. As in the FD case, “id” refers to a cross-section identifier.